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from collections import Counter
import nltk
import string
import matplotlib.pyplot as plt
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('stopwords')

# News Articles
articles = """
Artificial intelligence is transforming healthcare and education systems
AI and machine learning are the future of technology and innovation
Healthcare systems are improving with artificial intelligence solutions
Education technology is growing rapidly with AI based tools
"""

# 1. Convert to lowercase
articles = articles.lower()

# 2. Remove punctuation
for p in string.punctuation:
    articles = articles.replace(p, "")

# 3. Tokenize text
words = word_tokenize(articles)

# Remove stopwords
stop_words = set(stopwords.words('english'))
filtered_words = [word for word in words if word not in stop_words]

print("Tokens:")
print(filtered_words)

# 4. Word Frequency Distribution
freq = Counter(filtered_words)

print("\nWord Frequency Distribution:")
print(freq)

# 5. Vocabulary
vocabulary = set(filtered_words)

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print("\nVocabulary:")
print(vocabulary)
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print("\nVocabulary Size:", len(vocabulary))
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# 6. Top 5 Frequent Words
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print("\nTop 5 Frequent Words:")
for word, count in freq.most_common(5):
    print(word, ":", count)
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# 7. Type-Token Ratio (TTR)
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ttr = len(vocabulary) / len(filtered_words)
print("\nType Token Ratio =", round(ttr, 2))
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# 8. Dominant Topic
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dominant_topic = freq.most_common(1)[0]
print("\nMost Dominant Topic Keyword:")
print(dominant_topic[0], "appears", dominant_topic[1], "times")
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# 9. Bar Chart
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word_list = list(freq.keys())
count_list = list(freq.values())
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plt.figure(figsize=(10, 5))
plt.bar(word_list, count_list)
plt.title("Technology News Word Frequency")
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```